Unit: Unit 3: Functions

Topic: Definition of Relation and Function, Domain, Range, and Notations

Name: ________________

Date: ____________

Quiz (Habanero)

Q1. Is the relation $\{(x, y) : x^2 + y^2 = 4\}$ a function?
- (A) Yes
- (B) No
- (C) Maybe
- (D) Insufficient data
- (E) Cannot be determined

Q2. Given the function $f(x) = 2x + 1$, what is $f(3)$?
- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 9

Q3. What is the domain of the function $f(x) = \frac{1}{x}$?
- (A) All real numbers
- (B) All real numbers except 0
- (C) All integers
- (D) All rational numbers
- (E) All irrational numbers

Q4. Which of the following is a valid notation for the function $f(x) = x^2$?
- (A) $f(x) = x^2$
- (B) $g(x) = x^2$
- (C) $h(x) = x^2$
- (D) $f(y) = y^2$
- (E) All of the above

Q5. Given the graph of a function, what is the range of the function if the lowest point on the graph is $(-2, -3)$ and the highest point is $(4, 5)$?
- (A) $[-3, 5]$
- (B) $[-5, 3]$
- (C) $[-2, 4]$
- (D) $[-4, 2]$
- (E) $[-1, 1]$

Q6. The temperature in a city is given by the function $T(t) = 20 + 5\sin(\frac{t}{10})$, where t is time in hours. What is the temperature at $t = 30$ hours?
- (A) 20
- (B) 22
- (C) 25
- (D) 22.5
- (E) 21.25

Q7. What is the period of the function $f(x) = 3\sin(2x)$?
- (A) $\frac{2\pi}{3}$
- (B) $\frac{\pi}{2}$
- (C) π
- (D) $\frac{\pi}{3}$
- (E) 2π

Q8. The amount of rainfall in a region is given by the function $R(t) = 10 + 2t$, where t is time in months. What is the amount of rainfall after 5 months?
- (A) 10
- (B) 12
- (C) 15
- (D) 20
- (E) 14

Open Ended Questions (FRQ)

Q9. The height of a tree is given by the function $h(t) = 2t + 1$, where t is time in years. If the tree is currently 11 feet tall, how many years ago was it 5 feet tall? Show all work and explain your answer.

Q10. The population of a city is given by the function $P(t) = 1000 + 50t$, where t is time in years. If the population is currently 2000, what will be the population in 10 years? Show all work and explain your answer.

Chef's Tips

- Chef's Tips: Make sure to check for any restrictions on the domain of the function.
- Chef's Tips: Use the vertical line test to determine if a relation is a function.
- Chef's Tips: When evaluating a function, be sure to plug in the correct value for the input variable.